

Spatial Genomics / Spatial Transcriptomics

Keeping the map: measuring expression with position retained

Genomics Hackathon — methods primer

Contents

1	Where this material comes from	1
1.1	Recap: how we turn a tissue into numbers (from Lecture 1)	1
1.2	Single-cell RNA-seq, in one paragraph	1
2	Why position matters — what dissociation destroys	2
2.1	The dissociated view	2
2.2	The spatial view	2
2.3	What only the spatial view can show	2
3	The spatial methods and their trade-offs	2
4	What you can compute with spatial data	3
4.1	Reconstructing an axis → a zonation index	3
4.2	Niche / neighborhood analysis	3
4.3	Spatial ligand–receptor inference	4
5	Limitations — keep these honest	4
6	Connecting to this project	4

Sourcing note. This is a synthesis primer for the hackathon. Single-cell and sequencing foundations are grounded in *Genomics Lecture 1*. The spatial-transcriptomics content is compiled from standard methods knowledge plus the project’s own framing, because it is *not* present in the typed English lecture notes — those notes contain no dedicated spatial-transcriptomics lecture (the lecture called “spatial,” Lecture 12, is about Hi-C / 3D-genome folding, which is a different meaning of “spatial”). The single-cell recap below follows Lecture 1.

From dissociated single cells to expression with a street address — and using a spatial map as a ruler to decode disease. Builds on Genomics Lecture 1 (sequencing foundations); audience: students who know scRNA-seq basics.

1 Where this material comes from

1.1 Recap: how we turn a tissue into numbers (from Lecture 1)

Sequencing reads molecules in parallel. Sanger uses chain-terminating ddNTPs that stop synthesis randomly across many template copies, then sorts fragments by length. Illumina spreads fragments

onto a flow cell, amplifies each into a local cluster, and photographs one base per cycle (“sequencing by synthesis”). Crucially, the physical position of a cluster on the flow cell is meaningless — it just says “a read is here”; genomic identity comes only from alignment.

Hold onto that last point. In ordinary sequencing the (x, y) on the machine is throwaway. The whole idea of spatial transcriptomics is to make a position (x, y) mean something biological again — not on the flow cell, but in the tissue.

1.2 Single-cell RNA-seq, in one paragraph

RNA-seq measures gene expression by sequencing the mRNA in a sample and counting reads per gene. Single-cell RNA-seq (scRNA-seq) does this one cell at a time: the tissue is dissociated into a suspension, each cell is captured (e.g. in a droplet) with a bead carrying a unique **cell barcode**, every captured mRNA gets a **UMI** (unique molecular identifier) so PCR duplicates can be collapsed, and you end up with a cells $\times$ genes count matrix. Cluster that matrix and you recover cell types.

Key vocabulary. scRNA-seq = single-cell RNA-seq (whole cell). snRNA-seq = single-nucleus RNA-seq — only the nucleus is profiled, which is the trick that lets you measure frozen or fibrotic tissue where whole cells will not survive dissociation. Both are *dissociated* methods.

2 Why position matters — what dissociation destroys

There are two fundamentally different ways to measure a tissue, and the difference is entirely about whether you keep the cells’ addresses.

2.1 The dissociated view

To do scRNA-seq / snRNA-seq you must dissociate the tissue — chop and digest it until cells (or nuclei) float free in a suspension. The reward is enormous: you read essentially the whole transcriptome of each cell, thousands of genes, at single-cell resolution. The price is brutal and irreversible:

What is lost. The moment cells go into suspension, their position is destroyed. Cell #5,000 and cell #5,001 in your matrix might have been neighbors, or on opposite sides of the tissue — you can never tell. You get a perfect parts list with the assembly diagram thrown away.

2.2 The spatial view

Spatial transcriptomics measures expression *without* dissociating — on an intact tissue section, so each measurement carries its (x, y) coordinate. You keep the assembly diagram.

The core distinction. Dissociated = high transcriptome coverage, position gone. Spatial = expression tied to a location, usually with some cost in coverage or sensitivity.

2.3 What only the spatial view can show

- **Tissue architecture:** map each cell type onto where it actually sits — layers, structures, lesions.
- **Gradients & axes:** reconstruct how a gene rises or falls along a spatial direction. The textbook case is *liver zonation* — hepatocytes change their metabolic program continuously along the

porto-central axis (from the portal triad to the central vein). Dissociation flattens this gradient into noise; spatial data recovers it as a smooth curve.

- **Niches / neighborhoods:** ask which cell types sit next to which — e.g. immune cells clustering around a tumor edge.
- **Spatial ligand–receptor inference:** a signaling cell and a responding cell must be close; only with coordinates can you score whether a ligand-expressing cell actually neighbors a receptor-expressing one.

3 The spatial methods and their trade-offs

No method gives you everything. They sit on a triangle of competing goods: **transcriptome breadth** (how many genes), **spatial resolution** (how fine the (x, y)), and **sensitivity** (how many molecules you actually catch). Two broad families exist.

Sequencing-based (capture & sequence). The tissue is laid on a slide patterned with spatially-barcoded capture spots. mRNA diffuses down, binds a spot, and inherits that spot’s spatial barcode — the same barcoding trick as scRNA-seq, except the barcode now encodes a *tissue location*. Whole-transcriptome, but resolution is limited by spot size. (Notice the echo of Lecture 1: a barcode tells the sequencer where a read belongs — here “where” is a real place.)

Imaging-based (in situ hybridization / sequencing). Genes are detected directly inside the intact cell by fluorescent probes imaged under a microscope, often one molecule at a time. Subcellular resolution, but you can only target a pre-chosen panel of genes, not the whole transcriptome.

Method	Family	Resolution	Gene coverage	Note
10x Visium	Sequencing	Spots $\sim 55\ \mu\text{m}$ (a few cells each)	Whole transcriptome	Workhorse; spots are multi-cell, so not true single-cell.
Visium HD	Sequencing	2–8 μm grid	Whole transcriptome	Near single-cell on a continuous grid.
Slide-seq	Sequencing	$\sim 10\ \mu\text{m}$ beads	Whole transcriptome	Bead-array; high spatial detail, sparser per bead.
MERFISH	Imaging (ISH)	Subcellular	100s–1000s targeted genes	Single-molecule, combinatorial barcodes.
seqFISH	Imaging (ISH)	Subcellular	100s–1000s targeted genes	Sequential hybridization rounds.
In-situ sequencing	Imaging (ISS)	Subcellular	Targeted	Sequences short barcodes in place.
PhenoCycler / CODEX	Imaging (protein)	Subcellular	~ 40 –100 proteins	Measures protein, not RNA — complementary.

The one-line trade-off. Sequencing-based methods (Visium, Slide-seq) win on gene breadth but are coarser in space and often catch multiple cells per spot. Imaging-based methods (MERFISH, seqFISH) win on resolution and sensitivity but only for a targeted panel. Choose your corner of the triangle to match the question.

4 What you can compute with spatial data

4.1 Reconstructing an axis $\rightarrow$ a zonation index

Once every measurement has an (x, y) , expression becomes a function of space. If a tissue has a dominant geometric axis (liver’s porto-central axis, a cortical depth, a crypt–villus axis in gut), you can project each spot onto that axis and ask how a gene varies along it.

$$\text{zone-index}(\text{cell}) = \text{position of cell along the porto-central axis} \in [0, 1].$$

Concretely: pick a panel of genes known to be high at one pole and low at the other, combine them into a continuous score, and you get a quantitative *zonation index* — a single number per cell saying “how portal vs. how central am I.” This turns a vague anatomical notion into a measurable coordinate.

4.2 Niche / neighborhood analysis

For each cell, look at the cell types within a small radius and summarize that local mixture. Recurring mixtures define *niches* — reproducible micro-environments (e.g. “hepatocyte + endothelial + Kupffer”). You can then test whether a niche changes in disease.

4.3 Spatial ligand–receptor inference

Signaling needs proximity. With coordinates you score ligand–receptor pairs only where the partners are actually adjacent, distinguishing real local crosstalk from coincidence.

5 Limitations — keep these honest

Caveats.

- **Resolution vs. depth:** the more genes you read (sequencing-based), the coarser the spot; the finer the spot (imaging-based), the fewer genes — you rarely get both.
- **Multi-cell spots:** a Visium spot mixes a few cells, so its profile is a blend. Recovering per-cell composition needs computational deconvolution against a single-cell reference.
- **Dropout & sparsity:** lowly-expressed genes are missed (a zero may mean “absent” or merely “not captured”), and per-spot/per-bead counts can be shallow.
- **2D slices of 3D tissue:** a section is one plane; axes and niches are reconstructed from a sample, not the whole organ.

6 Connecting to this project

Here is the move at the heart of the hackathon, stated plainly.

The big idea: spatial as a ruler. A spatial dataset gives us a *reference map* in which expression is tied to position — for the liver, where each expression profile sits along the porto-central zonation axis. We learn, from that reference, the relationship “this expression pattern $\rightarrow$ this zone-index.”

Now take a dissociated dataset — snRNA-seq of disease tissue, where position was destroyed at the dissociation step. On its own it has no coordinates. But because the spatial reference taught

us the expression-to-position mapping, we can apply that mapping back onto the dissociated cells and decode an inferred position for each one.

spatial reference (learn)	→	expression $\Rightarrow$ zone-index
dissociated disease cells (apply)	→	recover lost position

The spatial map is the ruler; the dissociated data are the measurements we re-place onto it. snRNA-seq matters here precisely because the disease tissue is frozen/fibrotic — nuclei survive where whole cells would not. We trade the irreversible loss of position for the ability to profile hard tissue, and then buy the position back computationally using the spatial reference.

Why this is powerful. It lets us ask spatial questions — “is zonation disrupted in disease? which zone do the sick cells come from?” — on datasets that have no spatial information of their own. The spatial reference effectively donates its coordinate system to dissociated data.

One-screen summary

- **Dissociated (sc/snRNA-seq):** all genes, no position.
- **Spatial (Visium, MERFISH, ...):** position kept, trade-offs in breadth/resolution.
- With position you can reconstruct axes (zonation), niches, and local signaling.
- **This project:** use the spatial axis as a ruler to decode lost positions onto dissociated disease cells.

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